INTERNATIONAL TELECOMMUNICATION UNION

TELECOMMUNICATION STANDARDIZATION SECTOR

STUDY PERIOD 2017-2020

SG5-XXX-XXX

STUDY GROUP 5

Original: English

Question(s): 12/5 Sophia Antipolis, 13-23 June, 2023

CONTRIBUTION

Source:	India
Title:	New work item for the creation of a Supplement to Recommendation ITU-T L.1700 on the subject of Low-cost sustainable telecommunication for rural communications in developing countries enabling SIP based voice calling on WLAN/Wi-Fi.
Purpose:	Proposal

Contact:	Sandeep Agarwal CDOT Department of Telecommunications India	Tel: +91-80-2511-9850 E-mail: sandeepa@cdot.in;
Contact:	Neha Upadhyay Telecommunication Engineering Centre Department of Telecommunications India	Tel: +91 11 23323107 E-mail: neha.upadhyay86@gov.in

Keywords: New work item, Low-cost, sustainable telecommunication, rural communication, WLAN/Wi-Fi, SIP

Abstract: This contribution seeks to propose creation of a new Work Item under Question 12/5 for formulation of a new Supplement to Recommendation ITU-T L.1700 on the subject of Low-cost sustainable telecommunication for rural communications in developing countries enabling SIP based voice calling on WLAN/Wi-Fi

1. Introduction

Access to communication and ICT infrastructure in rural areas of most developing countries is still a challenge. The inadequacy of crucial infrastructure like electricity in rural and remote areas present a significant problem. Deployment of telecommunication solutions which are low-cost and low maintenance and can be deployed easily is an inevitability in such cases. WLAN/Wi-Fi network infrastructure, configured to provide traditional telecom services like voice calling, being an overall low cost solution can prove to be a viable solution in such cases to provide low cost sustainable telecommunication.

2. Proposal

This contribution proposes a new Work Item under Question 12/5 for formulation of a new Supplement to Recommendation ITU-T L.1700. The A.1 form is annexed with this document as Annexe I. A suggested base text for the proposed study is also enclosed as Annexe II with the contribution.

ANNEXE I

A.1 justification for proposed draft new Supplement with ITU-T Recommendation L.1700

<table>
<tr><td>Question:</td><td>12/5</td><td>Climate change and assessment of information and communication technology (ICT) in the framework of the Sustainable Development Goals (SDGs)</td><td colspan="2">Sophia Antipolis, 13-23 June, 2023</td></tr>
<tr><td>Reference and title:</td><td colspan="4">Supplement describing a low cost sustainable telecommunication solution in rural and remote areas using WLAN/Wi-Fi for conventional telecom services like voice calling.</td></tr>
<tr><td>Base text:</td><td colspan="2">Annex II

Base text for the proposed Supplement describing a low cost sustainable telecommunication solution in rural and remote areas using WLAN/Wi-Fi for conventional telecom services like voice calling</td><td>Timing:</td><td>Study Period 2022-2024</td></tr>
<tr><td>Editor(s):</td><td colspan="2">Neha Upadhyay
Tel: +91 11 23323107
E-mail: neha.upadhyay86@gov.in

Sandeep Agarwal
Tel: +91-80-2511-9850
E-mail: sandeepa@cdot.in;</td><td>Approval process:</td><td>AAP</td></tr>
<tr><td colspan="5">Scope
The scope of this work item is to describe a low cost sustainable telecommunication solution in rural and remote areas using WLAN/Wi-Fi for conventional telecom services like voice calling.</td></tr>
<tr><td colspan="5">Summary

The supplement provides description of low cost sustainable telecommunication solution using WLAN/Wi-Fi network infrastructure in rural and remote areas where crucial infrastructure like electricity etc. pose a major concern in deployment and sustenance of telecom infrastructure.</td></tr>
<tr><td colspan="5">Relations to ITU-T Recommendations or to other standards (approved or under development):

ITU-T L.1700</td></tr>
<tr><td colspan="5">Liaisons with other study groups or with other standards bodies:

ITU-D</td></tr>
<tr><td colspan="5">Supporting members that are committing to contributing actively to the work item:

India</td></tr>
</table>

ANNEXE II

Base text for the proposed Supplement describing a low cost sustainable telecommunication solution in rural and remote areas using WLAN/Wi-Fi for conventional telecom services like voice calling

1. Scope

This aims at identifying a low-cost sustainable solution based on WLAN/Wi-Fi network which can provide conventional telecommunication services like voice calling to other cellular as well as fixed line phones. This solution also utilises renewable sources of energy, and cost efficient backhaul in areas where optical fiber penetration is low or not present at all.

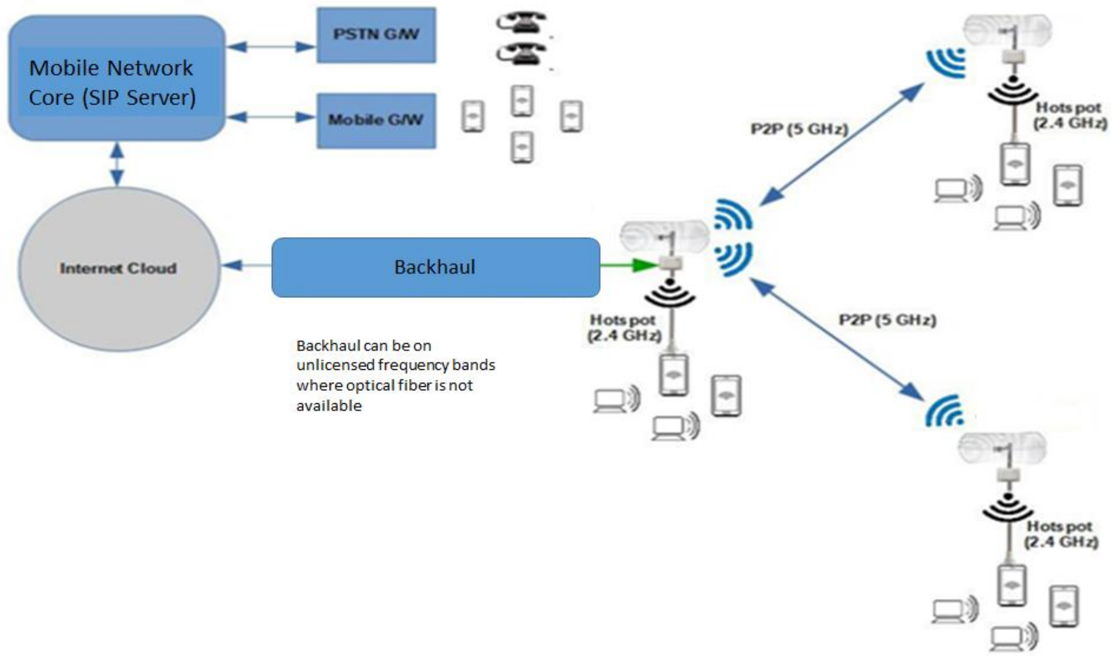
2. Introduction

This solution proposes use of WLAN/Wi-Fi network to facilitate SIP based calling to other mobile or landline networks through the core network of a conventional mobile network service provider. The WLAN/Wi-Fi Access Point (AP) used in this solution is a multi-radio platform. The same AP is used to provide Hotspot connectivity at 2.4 GHz as well as P2P/P2MP (Point to Point/ Point to Multi Point) configuration at 5 GHz by using suitable antennas.

WLAN/Wi-Fi access network at a location is provided by WLAN/Wi-Fi equipment in Hotspot, P2P & P2MP configuration. The user devices use SIP client to generate VoIP call over WLAN/Wi-Fi network which is directed to the Telecom Service Provider VoIP core and terminated at mobile/landline networks through respective gateways and vice-versa.

3. System Architecture

A typical system architecture for this solution is as below



4. Network components in the solution

A typical deployment of this solution has the following network components:

Sr. No.	Equipment	Details
1.	WLAN/Wi-Fi Access Points	Frequency of operation: 2.4 GHz, 5 GHz Mode of operation: P2P, P2MP and Hotspot configuration Technology: IEEE 802.11a/b/g/n/ac
2.	Antennas	Sector and long range antenna for access as well as P2P/P2MP links
3.	SIP client	Software for placing calls to mobile/landline networks through WLAN/Wi-Fi network
4.	Power supply	Solar panels and cells (Conventional power supply wherever available can be used)
5.	Backhaul connectivity	Can be based on • Backhaul using P2P or P2MP configuration in unlicensed bands of 5 GHz or 2.4 GHz of WLAN/Wi-Fi APs using long range antenna • Optical fiber wherever available • Conventional microwave backhaul wherever available
6.	Element Management System	To manage and configure the WLAN/Wi-Fi AP and other related network devices at remote locations
7.	Core network of Mobile Service provider	Mobile core network with SIP server or VoIP enabling core network components.

5. Overview of the solution

The solution aims to provide WLAN/Wi-Fi coverage in rural and remote areas where coverage by other conventional sources is limited or not present at all. A brief overview of the system is as below:

- *WLAN/Wi-Fi Access Point (APs) deployment and WLAN/Wi-Fi Coverage Provision*
  The WLAN/Wi-Fi APs will be deployed at different locations in a region where coverage has to be provided. The WLAN/Wi-Fi AP can be used with sector antennas instead of omnidirectional antennas to extend the coverage. One of the sectors of the WLAN/Wi-Fi AP can be used to extend P2P link to backhaul. The necessary AAA (Authentication, Authorization & Accounting) provisions are made to ensure necessary

AAA functionalities. FCAPS functionality for service provisioning for users and network equipment management is also deployed.

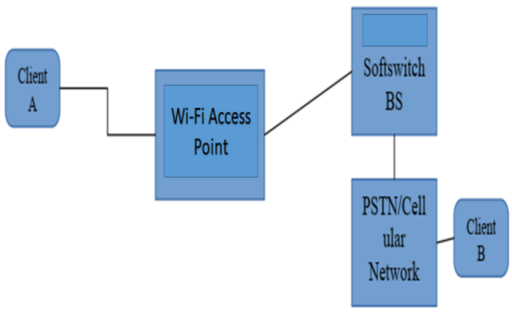
- *SIP client based voice calling over WLAN/Wi-Fi to mobile and landline network*s

SIP based calling through WLAN/Wi-Fi network can enable make or receive calls to and from mobile and landline networks. The call is routed through the SIP server installed in the core network of the anchor service provider (Mobile or fixed line service provider). The service provider will authenticate the user and provide him with a unique calling number (based on the relevant numbering scheme employed in the network) which will be entered in the SIP client to facilitate calling to other mobile or fixed line networks. Whenever, user will make a call through the SIP client, the called party will be displayed this number allocated to the user. The called party will receive the call as normal mobile or fixed-line telephone call and no SIP client or any other VoIP client needs to be installed at the user device of the called party. This facility will enable the users to make conventional calls using WLAN/Wi-Fi in rural and remote areas where coverage of mobile or landline networks is limited or not present at all. This will also curb the capacity planning issues of the mobile network service providers in certain areas as the traffic in this case is being offloaded over the WLAN/Wi-Fi network.

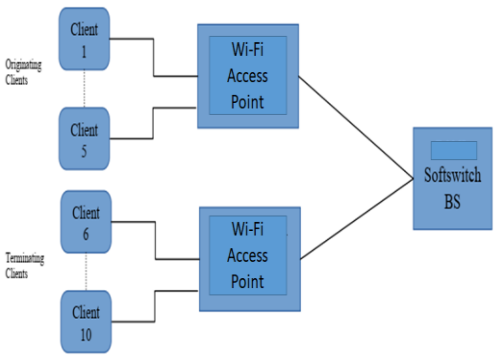
6. Different communication scenarios of the solution

The different communication scenarios of the solution include the following:

Scenario A



Scenario B



7. Features of the solution

i. ***Low cost:*** The solution offers conventional services like voice calling apart from data services over WLAN/Wi-Fi network which reduces the charges of licensed spectrum for providing access, mobile network equipment charges, backhaul spectrum charges etc. This will also curb the capacity planning issues of the mobile network service providers in certain areas as the traffic in this case is being offloaded over the WLAN/Wi-Fi network

ii. ***Solar based power supply:*** The whole WLAN/Wi-Fi AP system with sector based coverage and backhaul on a WLAN/Wi-Fi link as well can be powered by Solar based power supply which not only provides sustainability but also reduces the dependence of the network on conventional grid power supply thereby aiding in increasing the reliability of the solution in rural and remote areas.

iii. ***Easy deployment & management:*** WLAN/Wi-Fi networks are easily deployable due to their low degree of complexity as compared to other conventional telecommunication networks.

iv. ***User authentication:*** The user authentication for use of SIP client based calling can be facilitated by the anchor service provider. Different anchor service providers can provide services in different or same areas.

8. Summary & Conclusion

The solution proposed here offers a low cost and sustainable way of extending coverage in rural areas through WLAN/Wi-Fi deployment and provide conventional services like voice calling (using SIP client) alongwith the data services.

9. REFERENCES

i. http://www.dot.gov.in/
ii. http://www.tec.gov.in
iii. https://www.cdot.in